MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project MansourahMassarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION 49EE PROCESS PLANT SEWERAGE NETWORK DESIGN CALCULATION

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1. INTENT

This document is prepared for the design calculation of theTrunk Sewerage network for the process palnt area of Mansourah&Massarah (MMGP) Gold Project. This document includes sewerage discharge from different buildings for the above mentioned process Plant area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah&Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for the trunk sewerage network system of non plant buildings of process plant area.

Sewerage discharges from different non plant buildings like Engineering offices, Warehouse and Workshops, Ablution Buildings, Gold room, guard rooms etc are being connected through a common network with intermediate pumping stations to deliver the entire sewage water to a common sump (lifting station) for final discharge to the sewage treatment plant.

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- 80% of the supplied potable water (in litre / per head/ day) has been considered to be discharged at the sewage network.
- An additional 250 litre/ day / inspection pit (manholes) of rain water has been taken into account as an ingress to the sewage network.
- Design basis as indicated in Section 13 of document # MD-216-0000-EG-CV-CRT-0001 has been considered and the recommendation of SBC 701 has also been taken into account.

4. PEAK DISCHARGE FROM BUILDING

Peak Sewerage discharges (3 times the average discharge) from different non plant buildings like Engineering offices, Warehouse and Workshops, Ablution Buildings, Gold room, guard rooms etc are listed below Table.

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Sl. No.	Name of the Buildings	Number of Users (Per Day)	Additional 20% as floating users	Water consumption in litre/day (considering @ 300 litre / head/ day)	Average sewage water discharge in litre/sec (sewage generation @ 80% of supply water)	Peak Discharge in litre/sec
1	Ablution Building	120	144	43200	0.40	1.2
2	Engineering Office cum CCR	15	18	5400	0.05	0.15
3	Laboratory	20	24	7200	0.07	0.2
4	Warehouse & Workshop	20	24	7200	0.07	0.2
5	Power Plant-Guard Room	5	6	1800	0.02	0.05
6	Power Plant-Control Room/Sub-station	5	6	1800	0.02	0.05
7	Fuel Unloading area Control Room	5	6	1800	0.02	0.05
8	Fuel Central Control Room	5	6	1800	0.02	0.05
9	Emergency Service Building	35	42	12600	0.12	0.35
10	Mosque & Ablution	160	192	57600	0.53	1.6
11	Administrative building	80	96	28800	0.27	0.8
12	Security Support Building	5	6	1800	0.02	0.05
13	Gold Room	4	4.8	1440	0.01	0.04
Outside Process area						
1	Weighbridge Operator Room	2	2.4	720	0.01	0.02

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5. SYSTEM DESCRIPTION&BASIS OF DESIGN

EXTERNAL SANITARY SEWER AREA

Gold Room

Sanitary sewage from this building, consisting of Toilet Block with 2 WC, 3 wash Basin and 1 Shower is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to the lifting Station LS 1 after covering 7 nos of manholes on the way.

So sewage water quantity discharge from Gold room will be 0.04 litre / sec (peak discharge) and this will be delivered to the LS-1.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 7 manholes it will be 0.02 litre /sec.

So final loading on LS-1 will be 0.06 litre /sec

Provided minimum **150 mm dia. PVC pipe up to LS-1**with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

Route from LS-1 to LS-2

Ablution Building

Sanitary sewage from this building, consisting of 5 nos WC, 8 nos Urinal, 9 nos wash Basin and 6 nos of showers is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

Building sewer from Ablution Building 1.2 litre / sec

Engineering Office cum CCR

Sanitary sewage from this building, consisting of 2 nos WC, 1 no Urinal, 2 nos wash Basin is proposed to be collected from the inspection chamber manhole (MH-1 03A & B) and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

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Building sewer from Engineering office cum CCR is 0.15 litre/sec.

Laboratory

Sanitary sewage from this building, consisting of 3nos WC, 2nos Urinal, 5nos wash Basin is proposed to be collected from the inspection chamber manhole (MH-1 06A & B) and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

Building sewer from Laboratory is 0.2 litre per sec

All the above three building sewerage will be connected to the trunk sewerage route from LS-1 to LS-2.

This route will have around 13 nos of manholes.

Hence rain water ingress will be (250 LPD / manhole) 0.002 litre/ sec in each manhole. So from 13 manholes it will be 0.038 litre/ sec.

Discharge received from LS-1 is 0.06 litre/ sec

Hence total discharge at LS-2 is (0.06 + 1.2 + 0.15 + 0.2+0.038) 1.648 litre / sec

Provided minimum **150 mm dia. PVC pipe from LS-1 to LS-2** with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

Route from LS-2 to LS-3

Warehouse & Workshop

Sanitary sewage from this building, consisting of 5 nos WC, 6 nos wash Basin and 1 no shower is proposed to be collected from the inspection chamber manhole (MH-11A & B) and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

Building sewer from Laboratory is 0.2 litre per sec

Warehouse and Workshop building sewerage will be connected to the trunk sewerage route from LS-2 to LS-3.

This route will have around 10nos of manholes.

Hence rain water ingress will be (250 LPD / manhole) 0.002 litre/ sec in each manhole. So from 10 manholes it will be 0.028 litre/ sec.

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Discharge received from LS-2 is 1.648litre/ sec

Hence total discharge at LS-3 is (1.648 + 0.2+ 0.028) 1.876 litre / sec

Provided minimum **150 mm dia. PVC pipe from LS-2 to LS-3** with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

Route from LS-3 to LS-5

Guard Room

Sanitary sewage from this building, consisting of 1 no WC, 1 no wash Basin is proposed to be collected from the inspection chamber manhole (MH-19A) and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

Building sewer from Guard is 0.05litre per sec

Guard room sewerage will be connected to the trunk sewerage route from LS-3 to LS-5.This route will have around 8 nos of manholes.

Hence rain water ingress will be (250 LPD / manhole) 0.002 litre/ sec in each manhole. So from 8 manholes it will be 0.016 litre/ sec.

Discharge received from LS-2 is 1.876 litre/ sec

Hence total discharge at LS-5 is (1.876 + 0.05+ 0.016) 1.942 litre / sec

Provided **200 mm dia. PVC pipe from LS-3 to LS-5** with 1 in 150 slope and Discharge capacity 20.9 litre / sec with flow velocity 0.94 m/s for 2/3 pipe full.

POWER PLANT CONTROL ROOM

Sanitary sewage from this building, consisting of Toilet Block with 2 WC, 2 wash Basin is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to the lifting Station LS-4 after covering 8 nos of manholes on the way.

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So sewage water quantity discharge from Control room will be 0.05 litre / sec (peak discharge) and this will be delivered to the LS-4.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-4. For 8 manholes it will be 0.016 litre /sec.

So final loading on LS-4 will be 0.066 litre /sec

FUEL LOADING AREA CONTROL ROOM

Sanitary sewage from this building, consisting of Toilet Block with 2 WC, 2 wash Basin is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to the lifting Station LS 4 after covering 1 nos of manholes on the way.

So sewage water quantity discharge from Control room will be 0.05 litre / sec (peak discharge) and this will be delivered to the LS-4.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 1 manholes it will be 0.002 litre /sec.

So final loading on LS-4 will be 0.052 litre /sec

So net loading on LS -4 will be (0.052 + 0.066) 0.118 litre / sec

Provided minimum **150 mm dia. PVC pipe up to LS-4** with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

Route from LS-4 to LS-5

This route has no intermediate connection.

Total manholes in this route 4 nos.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 4 manholes it will be 0.0115 litre /sec.

So final loading on LS-5from LS-4 SIDE due to rainwater ingress will be 0.0115 litre /sec

So net loading on LS -5 from LS-4 side will be (0.118 + 0.0115) 0.1295 litre / sec

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So net loading on LS-5 from all the sides will be (0.1295 +1.942) 2.0715 litre / sec.

Provided minimum **150 mm dia. PVC pipe from LS-4 to LS-5** with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

MOSQUE & ABLUTION BLOCK

Sanitary sewage from this building, consisting of Toilet Block with 5 WC, 25 wash Basin is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to the emergency Service Building and finally to lifting Station LS 6 after covering 4 nos of manholes on the way.

So sewage water quantity discharge from Mosque & Ablution Block will be 1.6 litre / sec (peak discharge) and this will be delivered to the route of Emergency Service Building

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 4 manholes it will be 0.008 litre /sec.

So final loading on Emergency Service Building will be 1.608 litre /sec

EMERGENCY SERVICE BUILDING

Sanitary sewage from this building, consisting of Toilet Block with 5 WC, 5 Urinal, 10 wash Basin is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to the emergency Service Building and finally to lifting Station LS 6after covering 3nos of manholes on the way.

So sewage water quantity discharge from Emergency Service Building will be 0.35 litre / sec (peak discharge) and this will be delivered to the route of LS-6.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 3 manholes it will be 0.006 litre /sec.

So final loading on LS-6will be (1.608 + 0.35 + 0.006) 1.964 litre /sec

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Provided minimum **150 mm dia. PVC pipe up to LS-6** with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

Route from LS-5 to LS-6

This route has no intermediate connection.

Total manhole in this route 6 nos.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 6 manholes it will be 0.012litre /sec.

So final loading on LS-6 from LS-5 SIDE due to rainwater ingress will be 0.014 litre /sec

So Net loading on LS-6 from all the sides will be (1.964+ 2.0715 + 0.012) 4.0475litre / sec

This will be finally discharged to the final sewage lifting station to Sewage Treatment plant.

Provided **200 mm dia. PVC pipe from LS-5 to LS-6** with 1 in 150 slope and Discharge capacity 20.9 litre / sec with flow velocity 0.94 m/s for 2/3 pipe full.

ADMINISTRATIVE BUILDING

Sanitary sewage from this building, consisting of Toilet Block with 2 WC, 2 Urinal, 3 wash Basin is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to theFSLP after covering 1nos of manholes on the way.

So sewage water quantity discharge from Administrative Building will be 0.8 litre / sec (peak discharge) and this will be delivered to the route of FSLP

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 3 manholes it will be 0.002 litre /sec.

So final loading on LS-6will be (0.8 + 0.002) 0.802litre /sec

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SECURITY SUPPORT BUILDING

Sanitary sewage from this building, consisting of Toilet Block with 2 WC, 2 Urinal, 1 wash Basin is proposed to be collected from the inspection chamber manhole and will be connected to the manhole of plant sewerage system through gravity sewer distribution network.

This line will lead to FSLP after covering 4 nos of manholes on the way.

So sewage water quantity discharge from Security Support Building will be 0.05litre / sec (peak discharge) and this will be delivered to the route of FSLP

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 4 manholes it will be 0.008 litre /sec.

So final loading on LS-6will be (0.05 + 0.008) 0.058litre /sec

Provided minimum **150 mm dia. PVC pipe up to FSLP**(Final Sewarage Lifting Pit) with 1 in 100 slope and Discharge capacity 11.9 litre / sec with flow velocity 0.95 m/s for 2/3 pipe full.

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Route from LS-6 to FSLP (FINAL SEWERAGE LIFTING PIT)

This route has two intermediate connections.

Total manhole in this route 5 nos.

Additionally rainwater ingress (@ 250 LPD per manhole) of 0.002 litre/ sec will also be discharged to the LS-1. For 7 manholes it will be 0.010 litre /sec.

So final loading on LS-6 from LS-5 SIDE due to rainwater ingress will be 0.010 litre /sec

So Net loading on FSLP from all the sides will be (4.0475+ 0.802 + 0.058+ 0.010) 4.9175litre / sec

Provided **250 mm dia. PVC pipe from LS-6 to FSLP** with 1 in 150 slope and Discharge capacity 38.0litre / sec with flow velocity 1.10 m/s for 2/3 pipe full.

This will be finally discharged to the final sewage lifting station to Sewage Treatment plant.

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6. REFERENCES

This document is to be read in conjunction with following document and drawings

[1] Structural Engineering Design Criteria :MD-216-1000-EG-ST-CRT-1001-A02

[2] Civil-General, Design Criteria :MD-216-0000-EG-CV-CRT-0001-C02

[3] Civil-General, Engineering Manual :MD-101-SMEM-EG-CV-CRT-0001

[4] Drainage &Sewerage, Engineering Manual :MD-101-SMEM-EG-CV-SPC-0004

[5] Sewerage System Drawing No. : MD-216-49EE-EG-CV-DED-1001-A01

MD-216-49EE-EG-CV-DED-1002-A01

MD-216-49EE-EG-CV-DED-1003-A01

MD-216-49EE-EG-CV-DED-1004-A01

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ATTACHMENT 1 COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

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BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

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PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.:** 216
STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE: BECHTEL-WTRAN-004010, 31 July, 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 31 July, 2020	Comment by	EPC Response DATE: 14 August, 2020	Status	Remarks
			General Comments (cc)				
MD-216-49EE-EG-CV-CAL-1001-A01	49EE	1	/24/60/60 not equal.	Mr.Doron Golan Bechtel	Heading indicates that " water consumption in litre / sec (considering @ 300 litre / head / day) " and the calculation remains in order. In the next column it is "Average sewage water discharge in litre/ sec" where the earlier values are made in litre/ sec and then as per sewage generation norms it has been taken @ 80% of supply water. So the corresponding liter / sec value has been multiplied by 0.8 to obtain the discharged sewage water quantity.	Closed	